

- Vectors in underline, matrices in capital. Assume all matrices have real entries unless stated otherwise.
- For a vector  $\underline{x}$ , we denote by  $x_i$  the  $i^{\text{th}}$  entry of  $\underline{x}$ .
- The inequality  $A \geq 0$  means that the matrix  $A$  is positive semi definite. The inequality  $\underline{x} \geq 0$  means that all the entries of the vector  $\underline{x}$  are non negative.
- Most of these problems are adapted from [1, 2] and related resources on the web.
- You may use either of `CVXPY` (Python), `CVX.jl` (Julia) or `CVX` (matlab) to solve these problems.

1. *Learning a quadratic pseudo-metric from distance measurements.*

We are given a set of  $N$  pairs of points in  $\mathbf{R}^n$ ,  $x_1, \dots, x_N$  and  $y_1, \dots, y_N$ , together with a set of distances  $d_1, \dots, d_N > 0$ . The goal is to find (or estimate or learn) a quadratic pseudo-metric  $d$ ,

$$d(x, y) = ((x - y)^T P (x - y))^{1/2},$$

with  $P \in \mathbf{S}_+^n$ , which approximates the given distances, *i.e.*,  $d(x_i, y_i) \approx d_i$ . (The pseudo-metric  $d$  is a metric only when  $P \succ 0$ ; when  $P \succeq 0$  is singular, it is a pseudo-metric.)

To do this, we will choose  $P \in \mathbf{S}_+^n$  that minimizes the mean squared error objective

$$\frac{1}{N} \sum_{i=1}^N (d_i - d(x_i, y_i))^2.$$

- Explain how to find  $P$  using convex or quasiconvex optimization. If you cannot find an exact formulation (*i.e.*, one that is guaranteed to minimize the total squared error objective), give a formulation that approximately minimizes the given objective, subject to the constraints
- Carry out the method of part (a) with the data given in `quad_metric_data.jl`. The columns of the matrices  $\mathbf{X}$  and  $\mathbf{Y}$  are the points  $x_i$  and  $y_i$ ; the row vector  $\mathbf{d}$  gives the distances  $d_i$ . Give the optimal mean squared distance error.

The provided code also generates a test set, with `X_test`, `Y_test`, and `d_test`. Report the mean squared distance error on the test set (using the metric found using the data set above).

2. *Duality and theorems of alternatives.*

Consider the following linear optimization problem where  $A \in \mathbf{R}^{m \times n}$  and  $c \in \mathbf{R}^n$ :

$$\begin{aligned} \min_{\underline{x}} \quad & c^T \underline{x} \\ \text{s.t.} \quad & A \underline{x} \preceq 0 \end{aligned}$$

- Derive the dual of the optimization problem above.
- Using the result from part (a), prove the following theorem of strong alternatives. Show that exactly one of the following two systems has a solution, for any matrix  $A$  and vector  $c$ :

- $A \underline{x} \preceq 0, \quad c^T \underline{x} < 0$
- $A^T \underline{y} + c = 0, \quad \underline{y} \succeq 0$

3. *Duals of some penalty function approximation problems.*

Derive a Lagrange dual for the problem

$$\begin{aligned} \min_{x, r} \quad & \sum_{i=1}^m \phi(r_i) \\ \text{s.t.} \quad & r = Ax - b, \end{aligned}$$

using conjugate functions.

(Optional:) What is the form of duals for the following penalty functions  $\phi : \mathbf{R} \rightarrow \mathbf{R}$ ? The variables are  $x \in \mathbf{R}^n$ ,  $r \in \mathbf{R}^m$ .

(a) *Deadzone-linear penalty* (with deadzone width  $a = 1$ ),

$$\phi(u) = \begin{cases} 0 & |u| \leq 1 \\ |u| - 1 & |u| > 1. \end{cases}$$

(b) *Huber penalty* (with  $M = 1$ ),

$$\phi(u) = \begin{cases} u^2 & |u| \leq 1 \\ 2|u| - 1 & |u| > 1. \end{cases}$$

(c) *Log-barrier* (with limit  $a = 1$ ),

$$\phi(u) = -\log(1 - u^2), \quad \text{dom } \phi = (-1, 1).$$

(d) *Relative deviation from one*,

$$\phi(u) = \max\{u, 1/u\} = \begin{cases} u & u \geq 1 \\ 1/u & u \leq 1, \end{cases}$$

with  $\text{dom } \phi = \mathbf{R}_{++}$ .

4. We consider the following problem.

$$\min_{x_1, x_2} \quad x_1^2 + x_2^2 \tag{1}$$

$$\text{s.t.} \quad (x_1 - 1)^2 + (x_2 - 1)^2 \leq 1, \tag{2}$$

$$(x_1 - 1)^2 + (x_2 + 1)^2 \leq 1 \tag{3}$$

1. Find the optimal solution and value of this problem.
2. Write and solve the KKT equation for this problem.
3. Derive and solve the Lagrangian dual of this problem.
4. Do we have strong duality?

5. *Numerical perturbation analysis example.* Consider the quadratic program

$$\min_{x_1, x_2} \quad x_1^2 + 2x_2^2 - x_1x_2 - x_1$$

$$\text{s.t.} \quad x_1 + 2x_2 \leq u_1,$$

$$x_1 - 4x_2 \leq u_2,$$

$$5x_1 + 76x_2 \leq 1,$$

with variables  $x_1, x_2$ , and parameters  $u_1, u_2$ .

- (a) Solve this QP, for parameter values  $u_1 = -2, u_2 = -3$ , to find optimal primal variable values  $x_1^*$  and  $x_2^*$ , and optimal dual variable values  $\lambda_1^*, \lambda_2^*$  and  $\lambda_3^*$ . Let  $p^*$  denote the optimal objective value. Verify that the KKT conditions hold for the optimal primal and dual variables you found (within reasonable numerical accuracy).

*Hint:* Check the documentation or users' guides for `CVXPY/CVX.jl` to find out how to retrieve optimal dual variables.

- (b) We will now solve some perturbed versions of the QP, with

$$u_1 = -2 - \delta_1, \quad u_2 = -3 - \delta_2,$$

where  $\delta_1$  and  $\delta_2$  each take values from  $\{-0.1, 0, 0.1\}$ . (There are a total of nine such combinations, including the original problem with  $\delta_1 = \delta_2 = 0$ .) For each combination of  $\delta_1$  and  $\delta_2$ , make a prediction  $p_{\text{pred}}^*$  of the optimal value of the perturbed QP, and compare it to  $p_{\text{exact}}^*$ , the exact optimal value of the perturbed QP (obtained by solving the perturbed QP). Recall the prediction  $p_{\text{pred}}^* = p^* + \delta_1 \lambda_1^* + \delta_2 \lambda_2^*$ .

Put your results in the two righthand columns in a table with the form shown below. Check that the inequality  $p_{\text{pred}}^* \leq p_{\text{exact}}^*$  holds. Why should this inequality hold?

| $\delta_1$ | $\delta_2$ | $p_{\text{pred}}^*$ | $p_{\text{exact}}^*$ |
|------------|------------|---------------------|----------------------|
| 0          | 0          |                     |                      |
| 0          | -0.1       |                     |                      |
| 0          | 0.1        |                     |                      |
| -0.1       | 0          |                     |                      |
| -0.1       | -0.1       |                     |                      |
| -0.1       | 0.1        |                     |                      |
| 0.1        | 0          |                     |                      |
| 0.1        | -0.1       |                     |                      |
| 0.1        | 0.1        |                     |                      |

6. *Projection on the  $\ell_1$  ball.* Consider the problem of projecting a point  $a \in \mathbf{R}^n$  on the unit ball in  $\ell_1$ -norm:

$$\begin{aligned} \min_x \quad & (1/2) \|x - a\|_2^2 \\ \text{s.t.} \quad & \|x\|_1 \leq 1. \end{aligned}$$

Derive the dual problem and describe an efficient method for solving it. Explain how you can obtain the optimal  $x$  from the solution of the dual problem.

7. *Optional, Source localization from range measurements* [Beck, Stoica, and Li, from B&V]. A signal emitted by a source at an unknown position  $x \in \mathbf{R}^n$  ( $n = 2$  or  $n = 3$ ) is received by  $m$  sensors at known positions  $y_1, \dots, y_m \in \mathbf{R}^n$ . From the strength of the received signals, we can obtain noisy estimates  $d_k$  of the distances  $\|x - y_k\|_2$ . We are interested in estimating the source position  $x$  based on the measured distances  $d_k$ .

In the following problem the error between the squares of the actual and observed distances is minimized:

$$\text{minimize} \quad f_0(x) = \sum_{k=1}^m (\|x - y_k\|_2^2 - d_k^2)^2.$$

Introducing a new variable  $t = x^T x$ , we can express this as

$$\begin{aligned}
& \min_{x, t} \quad \sum_{k=1}^m (t - 2y_k^T x + \|y_k\|_2^2 - d_k^2)^2 \\
& \text{s.t.} \quad x^T x - t = 0.
\end{aligned} \tag{1}$$

The variables are  $x \in \mathbf{R}^n$ ,  $t \in \mathbf{R}$ . Although this problem is not convex, it can be shown that strong duality holds. You do not have to show this. This is a variation on the problem discussed on page 229 and in exercise 5.29 of *Convex Optimization*.

Solve (1) for an example with  $m = 5$ ,

$$y_1 = \begin{bmatrix} 1.8 \\ 2.5 \end{bmatrix}, \quad y_2 = \begin{bmatrix} 2.0 \\ 1.7 \end{bmatrix}, \quad y_3 = \begin{bmatrix} 1.5 \\ 1.5 \end{bmatrix}, \quad y_4 = \begin{bmatrix} 1.5 \\ 2.0 \end{bmatrix}, \quad y_5 = \begin{bmatrix} 2.5 \\ 1.5 \end{bmatrix},$$

and

$$d = (2.00, 1.24, 0.59, 1.31, 1.44).$$

To solve the problem, you can note that  $x^*$  is easily obtained from the KKT conditions for (1) if the optimal multiplier  $\nu^*$  for the equality constraint is known. Derive the dual problem, express it as an SDP, and solve it using CVX to find  $\nu^*$ , and hence,  $x^*$ .

## References

- [1] Stephen Boyd and Lieven Vandenberghe. *Convex optimization*. Cambridge university press, 2004.
- [2] Stephen Boyd and Lieven Vandenberghe. Additional exercises for convex optimization, 2010.